A PROJECT TITLE

ON

Water Quality Classification Using SVM and Random Forest

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF Computer Science and Information Technology

Name of Title: Water Quality Classification Using SVM and Random Forest

1. Abstract:

Water quality plays a vital role in protecting human health, agriculture, industries, and the environment. Monitoring water quality is essential because contaminated water can cause serious health problems and affect ecosystems. Traditional methods of testing water quality require laboratory analysis, which can be time-consuming and expensive. Machine Learning (ML) provides a faster and more efficient way to analyze water quality by learning patterns from existing data and predicting the quality of water automatically.

This project focuses on Water Quality Classification Using Support Vector Machine (SVM) and Random Forest, two popular Machine Learning algorithms. The project uses a dataset containing important water quality parameters such as pH, hardness, dissolved solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity. These parameters are used as input features to train the ML models. Before training, the dataset is cleaned, preprocessed, and divided into training and testing sets to ensure accurate model evaluation.

The SVM algorithm classifies water quality by finding the best boundary between different classes, while the Random Forest algorithm combines multiple decision trees to improve prediction accuracy and reduce errors. After training, both models are tested on unseen data, and their performance is evaluated using accuracy and other evaluation metrics. The results are then compared to determine which algorithm provides better classification performance.

The main objective of this project is to develop an accurate and reliable model for classifying water quality and to compare the effectiveness of SVM and Random Forest algorithms. This project demonstrates how Machine Learning can simplify water quality analysis, reduce the time required for prediction, and support better decision-making for water resource management. It also helps students understand the practical application of Machine Learning techniques in solving real-world environmental problems.

2. INTRODUCTION:

Water quality is essential for human health, agriculture, industries, and the environment. Monitoring water quality helps identify whether water is safe for drinking or other purposes. Traditional testing methods can be time-consuming and expensive. Machine Learning provides a faster and more efficient approach to classify water quality using available data.

In this project, Support Vector Machine (SVM) and Random Forest algorithms are used to classify water as potable (safe) or non-potable (unsafe) based on water quality parameters such as pH, hardness, dissolved solids, sulfate, conductivity, and turbidity. The performance of both algorithms is compared to determine which model provides better prediction accuracy. This project demonstrates how Machine Learning can be applied to solve real-world environmental problems.

3. SOFTWARE REQUIREMENTS:

The following software is required to develop the Water Quality Classification Using SVM and Random Forest project:

- Operating System: Windows 10/11, Linux, or macOS
- Programming Language: Python 3.x
- IDE/Editor: Jupyter Notebook, Google Colab, or Visual Studio Code (VS Code)
- Libraries:
    - Pandas
    - NumPy
    - Matplotlib
    - Seaborn
    - Scikit-learn (sklearn)
- Dataset: Water Potability Dataset (CSV format)

Signature of the Guide

Course Instructor

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